

Understanding the Behavior of Overseas Debt Issuance by Non-financial Firms

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Abstract

This paper studies the behavior of the non-financial firms' overseas debt issuance using a newly developed database on international debt securities by the Bank of International Settlements. Our results suggest that depreciation of domestic currency against US dollar is negatively correlated with the international debt issuance. Domestic stock market growth rate is positively correlated with international debt issuance behavior in developed countries. Unlike developed countries, we find that exchange rate volatility plays a significant role in international debt issuance in the developing countries. These results are consistent with carry trade and currency mismatch hypothesis.

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1 Introduction

One of the puzzles during the 2013 episode of "Taper Tantrum" was the disproportionate response of financial markets in many emerging economies to expected tightening of monetary policy in the United States even though these countries had small net external debt position (measured in the usual residence terms) . Shin (2013) argued that one piece in the puzzle may be the role of non-financial firms that operate across borders. When corporate activity straddles the border, measuring exposures at the border itself may not capture the strains on corporate balance sheets, therefore net external debt position of a country is not a good indicator of its vulnerability to unfavorable external shocks. To take this problem into account, it has been suggested to focus on overseas debt issuance of non-financial firms (NFC hereafter) based on nationality. The relative importance of NFC's overseas debt issuance can be ascertained from the fact that more than half of the net "external" financing of emerging economies in 2012 took place through the issuance of international debt securities (Turner 2014). The relative importance of the non-financial firms' (NFC hereafter) overseas debt issuance in the recent periods has also been referred to as the second phase of Global Liquidity.

Until recently, not much attention has been paid by the academic world to NFC's overseas debt issuance. Partially it is due to the fact that financial sectors, instead of NFCs, issued the vast majority of international debt securities. Recently, more specifically, after 2009, we see more and more NFCs go outside their headquarter countries to issue foreign currency or domestic currency denominated debts in global financial market. During 1993-2015, the average growth rate of NFC international debt is 7% annually, which draws more attention from researchers to analyze this new phenomenon.

Given its importance for the stability of the global financial system, it is important to understand the behavior and determinants of NFC's overseas debt issuance. To do so, we utilize a recently developed database on international debt securities by the Bank of International Settlements. We perform a panel study of 32 countries for 1993-2015 sample period. Our paper improves on the

existing literature on international debt securities in three dimensions. One of the important issues in the study of overseas debt issuance is the transmission of international financial risk. Some researchers have argued that overseas debt issuance is no longer a serious problem for financial stability, as currency mismatch can be resolved through currency risk hedging instruments like exchange rate derivatives and currency swap contracts. Our paper examines this hypothesis by investigating the relationship between exchange rate and overseas debt issuance. In addition, our paper also contributes to the recent discussion about global liquidity. This strand of literature (Turner (2014) among others) suggests that global liquidity has become a relatively important factor in explaining the macroeconomic conditions and financial conditions. Shin (2013) argues that the second phase of global liquidity is driven by the overseas debt issuance by NFCs.

The third contribution of our paper relates to the use of debt issuance data based on nationality unlike most of the existing research which is based on residency. Several recent studies (Chung et al. (2014), Shin (2013)) argue that the residency-based categorization of debt issuance may not be able to capture some important features in international debt issuance. For example, an Indian corporate can defray its group cost in dollars by borrowing through its New York subsidiary. The firm then deposit Rupees in the local branch of the bank as a collateral. In this case, the firm has engineered a currency mismatch. In effect, the firm has taken on a carry trade position, holding cash in local currency financed with dollar liabilities.¹ From this perspective, our study provides a new way to look at NFC's international debt issuance activities or put it differently, serves as a complement to the existing residency-based studies.

Our results show that there is significant difference in the behavior of overseas debt issuance of NFCs of developed and developing economies. For developing economies, we find significant negative impact of level and volatility of exchange rate on changes in overseas debt issuance. For developed countries, the exchange rate volatility has no significant impact on the net debt issuance. The exchange rate level, however, shows a significant negative impact on net debt issuance after controlling for both the country and the year fixed effects. The significant relationship between

¹We will discuss the difference between these two types of categorization in details in the second part of this paper

exchange rate changes and overseas debt issuance supports the currency mismatch hypothesis that has been proposed in the literature. Other than domestic financial infrastructure, exchange regimes may also play a role in how debt issuance responds to exchange rate volatility. In case of floating exchange rate regime, the impact of volatility on the international debt issuance may be smaller, because investors may perceive the exchange rate fluctuation in these currencies as the natural outcome of market supply and demand conditions. On the other hand, if currencies under managed exchange rate regime suddenly become more volatile against major currency, it may imply hidden currency risks. Therefore, sophisticated international investors will be more concerned about investing in these countries. Our findings also suggest positive and significant relationship between stock return and overseas debt issuance. This relationship is stronger for developed economies. We also find that change in current account has a positive and significant impact on net debt issuance in developing countries implying that firms in developing countries may try to hedge the currency risk of its export earnings by issuing international debt securities in foreign currency.

The remainder of this paper is structured as follows. Section 2 reviews the literature on international debt securities. Section 3 presents our conceptual framework and econometric methodology followed by a discussion. In section 4 we discuss the results and section 5 concludes.

2 Literature Review

As mentioned before, this study uses nationality-categorized international debt issuance data to take into account the problems associated with the debt issuance data based on residence. This is motivated by recent global liquidity literature that also focuses on the behavior of debt issuance by NFCs based on nationality. Chung et al. (2014) explain the difference between two types of categorization: nationality-based and residency-based NFCs and why this difference is important in international debt issuance. The nationality-based NFC international debt is categorized by the headquarter country, no matter where the debt-issuing subsidiaries are located; while the

residency-based NFC's international debt is categorized by the debt-issuing subsidiary location. When a multinational firm borrows money through its overseas subsidiary from the international corporate bond market, the proceeds of the borrowing can either be sent to headquarters directly or through day-to-day business operation. In the former, the transaction is explicitly recorded as a capital flow item on the balance of payment; however, this is not the case in the latter. The residency-based categorization won't be able to capture the intra-corporation borrowing through the latter form. In the recent years, due to relatively strict regulation in the global banking sector, a new phenomenon has appeared in the international debt market, as Shin (2013) points out that in this second phase of global liquidity after Great Recession: many multinational firms borrow money from overseas subsidiaries, which issue the overseas debts, through day-to-day business operation, to avoid the international regulation of capital flow. The nationality-based categorization can solve this problem by consolidating the balance of payment between headquarter and overseas subsidy. Furthermore, we believe that, international debt issuance based on nationality-based categorization also captures some of the international financial risks that may be masked in the residency-based data.

Besides the importance of categorization standard, it is interesting to distinguish the heterogeneity between NFCs headquartered in developed countries and developing countries. For these NFCs headquartered in developed countries, these firms on average have a longer history of multinational business operation, i.e., they may have factories outside their home countries, therefore, the money they raise from international capital market does not necessarily need to be sent back to headquarter to pay for expenses. This is not true for developing countries. Those firms headquartered in developing countries are overall younger than the counterpart in developed countries; therefore they may only have foreign sales instead of foreign direct investment. If this is the case, they may borrow money overseas and send back to headquarter to pay for the expenses in domestic production. We conjecture that there exists more exchange rate risk from domestic currency against US dollar in developing countries than in developed countries.

Our study also sheds light on the currency mismatch literature in the context of international bor-

rowing. The debate about currency mismatch exists for a long time in the international finance literature. Recently, some researchers have suggested that currency mismatch is no longer an issue in international debt issuance, because of availability of currency derivatives and currency swap contracts. These financial instruments are used more and more to hedge the exchange rate risk generated by currency mismatch. However, there is still an open debate whether these financial derivatives do not completely hedge the exchange rate risk. This can be due to the fact that we are using nationality-categorized NFCs instead of residency. There is also evidence in the recent literature to explain why the exchange rate risk is not completely hedged. Chui et al. (2014) point out that many popular exchange rate derivatives have the KIKO feature, which is short for 'Knock-in, Knock-out'. These derivatives come with a call option (knock-in) held by the bank. If the reference currency (eg. the US dollar) strengthens beyond a certain threshold, the knock-in requires the exporter to sell its dollars at the strike price (i.e. below market rates and knock-out feature dictates that no option can be exercised by either the exporter or the bank if the dollar weakens below a certain threshold. These features save the hedge expenses, however, retain the tail risk of strong currency depreciations. Therefore, even with the presence of currency derivatives, the exchange rate risk may not be fully insured due to the features of these contracts. Currency swaps, another popular way to hedge exchange rate risk, depends on the availability of a swap counterparty. (Black and Munro, 2014). The swap counterparty is typically a non-resident issuing domestic currency debt such as World Bank (but generally with no use for domestic currency funding). If the swap counterparties are not available in a large scale, then this method may have a limited capability to hedge the exchange rate risk. To summarize, our paper also examines the currency mismatch hypothesis in the context of international debt securities. To do so, we can examine the relationship between net debt issuance by NFCs and exchange rate. In case of perfect hedging, we should not find any significant relationship between net debt issuance and exchange rate.

Carry trade may be another way to transmit international financial risks besides currency mismatch. NFCs may borrow money from overseas and speculate in the domestic stock market when it is booming. However, the capability to do so is constrained by the international capital flow reg-

ulation. If capital cannot freely flow across borders, then carry trade will be hard to implement. We conjecture that NFCs headquartered in developing countries may have a limited ability to conduct carry trade because of the international capital flow barrier, but not for most of the NFCs headquartered in developed countries where capital can flow more easily across borders.

3 Empirical Methodology and Data Description

Our study is based on a new database on international debt securities that have recently been developed by the Bank of International Settlements. We use data from 32 countries² during 1993-2015 based on data availability. The dataset comes from various sources, as listed in Table 1. The empirical model specification is as follows:

$$\begin{aligned} NetDebtIssuance_{it} = & \beta_0 + \beta_1 Exchange_Volatility_{it} + \beta_2 Exchange_Rate_{it} + \beta_3 GDP_Growth_{it} \\ & + \beta_4 Stock_Growth_{it} + \beta_5 \Delta CA_{it} + \beta_6 interest_gap_{it} + time_dummies + \varepsilon_{it} \end{aligned} \quad (1)$$

This model is estimated using fixed-effect panel regression with Newey-West robust standard errors. The description of the econometric methodology as well as data is provided below.

Net Debt Issuance: We first examine the stationarity property of overseas debt issuance by NFCs for each country. We could not reject the null of unit root at any conventional significance level. For sake of brevity, we don't report the results here. To take into this feature of the data, we take the first difference of NFC international debts (nationality based) to get the net issuance of debts for each quarter during the sample period. This variable measures the quantity of new debts issued in this quarter minus the quantity of existing debts paid off in this quarter, which gives the idea how much net debt issued each quarter. This variable includes all maturities of NFC international

²The 32 countries we use in this study are based on data availability. These countries are Argentina, Australia, Brazil, Canada, Chile, China, Czech Republic, Denmark, Hong Kong, Hungary, Iceland, India, Indonesia, Israel, Jamaica, Japan, Korea, Malaysia, Mexico, New Zealand, Norway, Peru, Philippines, Poland, Russia, Singapore, South Africa, Sweden, Switzerland, Thailand, Turkey, UK.

debts and is measured in terms of billions of US dollars. We apply seasonal adjustment to the level of debts before taking the first difference to correct for seasonality in the variable. The reason to use the first difference instead of the growth rate of debt is to avoid the inaccurate measure of the log difference of debt. Because if the initial level of debt stock is low, then after taking the first difference, the number can go to positive infinity if we subtract a term close to negative infinity in the first differencing.

Exchange Rate Level: We take the average of the daily closing exchange rate against US dollar within the quarter to measure the exchange rate level. The exchange rate is always measured using direct quotation, i.e. the exchange rate can be interpreted as how much the local currency one US dollar can purchase.

Exchange Rate Volatility: The exchange rate volatility is measured by the standard deviation of the daily closing exchange rate against US dollar within the quarter.

Stock Market Growth Rate: This variable is constructed to measure the growth rate of domestic stock market value. We calculate the growth rate from previous period of total share prices for all shares in the domestic stock market. FRED provides of total share prices for all shares for many countries, but not all of the 32 countries we study. When this variable is not available for some country, we use the growth rate from previous period of major stock market index instead. One should interpret this variable as quarterly stock market growth rate, i.e., the annual rate of stock market growth rate is around four times of this variable.

GDP Growth Rate: This variable measures the growth rate of real GDP from previous period for each country. We remove the seasonality in the quarterly GDP data first and take the first log difference of the seasonality-adjusted variable.

Change in Current Account Balance: Current account includes balance of trade, net primary income or factor income and net cash transfers, that have taken place over a given period of time. Since balance of trade is normally the largest component in current account, the change of current account balance is normally considered to reflect the trade surplus or deficit of the country. If the change of current account balance is positive, it means export increases more than import, then the exporting firms can expect to get paid by more foreign currency in the near future; if the change of current account is negative, it means import increases more than export, then exporting firms will receive less foreign currency payment. In corporate finance literatures, researchers turn to believe that NFCs normally do not speculate in the foreign exchange market. If NFCs have business involving foreign currency payment in the future, they should hedge the foreign exchange risks correspondingly. Theoretically we would predict, if the change in current account is positive, NFCs would expect to receive foreign currency payment in the future, so they should hedge the exchange rate risks by borrowing more foreign currency denominated debts now. By doing this, their foreign liability is hedged by the incoming foreign assets. Therefore, we expect to see a positive relationship between the change in current account and the change in net debt issuance. Here, we make seasonal adjustment to net current account variable (excluding exceptional financing) from International Financial Statistics (IFS) and then first difference the seasonal-adjusted variable, to measure the change in current account balance.

Interest Rate Gap: This variable is constructed using the domestic discount rate minus the federal fund rate in each quarter. NFCs may want to borrow from international financial market because of the interest rate gap between domestic market and the foreign market. The interest rate gap measures the price difference of using capital between domestic and foreign markets. Therefore, this variable is jointly determined by the supply and demand of the financial market. We are not clear about the coefficient sign of this variable since we do not have good controls for investors' characteristics.

The summary statistics are in Table 2. The average of net debt issuance is around 358 million US

dollars each quarter. The standard deviation of this variable is more than 2 billion US dollars, which reflects that there is a huge difference across countries and times. The maximum and minimum of net issuance in a quarter are 23.02 and -22.935 billion US dollars correspondingly. The panel unit root test results suggest there is no unit root in the net debt issuance, i.e. the panel regression should not suffer from spurious regression problem.

The exchange rate and exchange rate volatility both have large standard deviation compared to the corresponding mean. It is natural for exchange rate itself because each country issues local currency that does not need to have the same exchange rate against US dollar with another country. However, the exchange rate volatility differs significantly across countries and seems to have an asymmetric distribution in our dataset. The maximum of exchange volatility is much further away from the mean compared to the minimum, which indicates the distribution is right skewed, i.e., there are a few observations that have extremely large exchange volatility. This phenomenon reflects that there may be some historical periods when exchange rate volatility is larger than what it normally is, for instance, global financial crisis, or some countries may have suffered local currency crisis. To solve this problem, we use panel fixed-effect estimators and control for year-fixed effects to prevent the results to be driven by these extreme cases.

The average stock growth rate from previous quarter is 3.4%, which means on average we can expect the stock market return to be around 13% annually. This number is consistent with stock return literatures. The average real GDP growth rate from previous quarter is about 1%, so it implies 4% growth rate annually on average. The standard deviation is 1.8%, which is much smaller than the all other variables in this study. It confirms the common sense that real variables vary less than financial variables.

The average interest rate gap is around 5.5% annually. This positive gap suggests in the sample period, NFCs face cheaper financial costs overseas. If NFCs are sensitive to the cost of capital, they may try to borrow more from overseas. The summary statistics of change in current account is contaminated by extrapolation. China is an outlier in the sample since the current account balance is so large that it is completely in different quantity scale compared to other countries in the sample.

In this study, we will exclude China from the sample to perform a robustness check for the results.

4 Interpretation of Empirical Results

4.1 Financial risks associated with NFCs' overseas debt issuance

The fixed-effect panel regression results are shown in Table 3. These regressions use all the 32 countries during the sample period 1993-2015. Comparing the coefficients across all the regressions, the signs of coefficients and the significance of these coefficients have a high degree of robustness across different model specifications.

As shown in Table 3, volatility of exchange rate has a significant impact on net debt issuance across countries . The coefficients of exchange rate volatility and exchange rate level are both negative. The coefficients of exchange rate volatility are always negatively significant at one percentage level. A natural question arises in this context is why is there a negative correlation between exchange volatility and net debt issuance. It is possible that NFCs are risk-adverse agents in the financial market, so they may borrow less from international debt market if the exchange volatility is large. This is from the NFCs' perspective; how about from the international investors perspective? The investors are the agents who hold the major currency and want to purchase international debt securities for financial returns. When the major currency is appreciating against the local currency in the NFCs headquartered country, investors may have a concern about the NFCs ability to pay back the debts denoted in major currency with domestic business operation profits that are generally denominated in domestic currency units. It is hard to tell which one is the source of this negative relationship between these two variables; however, it is not difficult to draw the conclusion that currency mismatch still matters in international debt issuance from our results. Either NFCs or international investors, or both do take into account the exchange rate risks associated with NFCs' overseas debt issuance, which leads to a significant negative relationship between these two variables.

There is also some evidence in the relationship between exchange rate level and the net debt is-

suance to support the currency mismatch hypothesis. The coefficients of exchange rate level are negative with different degrees of significance. It implies when the local currency is depreciating, i.e., the direct quotation of exchange rate is larger, then the net issuance is decreasing, and vice versa. If the local currency is depreciating, the NFCs will bear more pressure to repay the international debt securities denominated in the foreign currencies; on the other hand, international investors will suspect if the NFCs have enough foreign liquidity to repay the international debt. Therefore we suggest that currency mismatch seems to play a role in the NFCs international debt issuance.

If NFCs do not have interest in speculating in foreign exchange market, why don't they hedge their exposure in that case? Exchange derivatives or currency swap seems to be a good option to deal with currency mismatch problems in practice. However, the ability to swap foreign currency funding into domestic currency is very often constrained by the supply of swap counterparty. (Black and Munro, 2013) If there is not enough swap counterparty domestic currency exposure, NFCs can only rely on foreign exchange derivatives to hedge the exchange rate risk. During the period when the exchange rate is volatile, it can be very costly to purchase a hedging contract to transfer the currency risk to a third party, therefore the potential cost to issue overseas can be much higher during this period. The NFCs may choose to under insure their exchange risks by hedging less than the exposures or purchasing KIKO feature derivatives.

To summarize our results so far, we find that currency mismatch plays a significant role in international debt securities market. If more risk appears in the foreign exchange market, the ability to issue more debt overseas may be constrained. This relationship could contribute a potential risk transmission channel in the global financial market. If NFCs need to rollover the existing debt by issuing new debt overseas, it could be hard for them to do so because the high exchange volatility may reduce the willingness for the international investors to purchase these debt securities or the NFCs' shareholders may hesitate to take more risk in the foreign exchange market. If they decide to default on these existing international debts, the financial risks can be transferred from NFCs' headquarter counties to all over the world.

Nevertheless, currency mismatch problems may differ across countries. We suspect that if the availability of foreign exchange derivatives differs across countries, then the currency mismatch issues may have different impact on international debt issuance. Therefore, we divide our sample into developed countries and emerging markets to see how different level of domestic financial development affects the relationship between the exchange rate volatility and net debt issuance. We report the results for developed countries in Table 4 and emerging markets in Table 5.

For developed countries, the exchange rate volatility has no significant impact on the net debt issuance after controlling for the exchange rate level in all model specifications. The exchange rate level, however, shows a significant negative impact on net debt issuance after controlling for both the country and the year fixed effects. This suggest that the currency mismatch still has the first-order impact on net debt issuance but lose the second-order impact. This is consistent with the fact that in developed countries, there may exist more financial instruments to hedge the risks in foreign exchange market. International investors and domestic NFCs are still sensitive to the impact of the exchange rate appreciation/depreciation on debt sustainability, but the second-order risks from exchange market fluctuation may be hedged by the NFCs headquartered in developed countries through the domestic sophisticated financial institutions.

For the emerging market economies, the results preserve the whole sample regression features in the coefficients of exchange rate level and exchange volatility. The impact of depreciation in local currency on net debt issuance is negatively significant and the impact of high exchange rate volatility against major currency is also negatively associated with net debt issuance. The contradictory results in the developed and developing countries suggest that there could exist some cross-country differences that explain why compared to developed economies, emerging markets are more sensitive to the second-order risks in exchange rate.

Other than domestic financial infrastructure, exchange regimes may also play a role in the cross-country differences. In case of floating exchange rate regime, the impact of volatility on the international debt issuance may be smaller, because investors may perceive the exchange rate fluctuation in these currencies as the natural outcome of market supply and demand conditions. On the other

hand, if currencies under fixed exchange rate regime suddenly become more volatile against major currency, it may imply hidden currency risks. Therefore, sophisticated international investors will be more concerned about investing in these countries. Roughly speaking, most of the developed countries have more float exchange rate regimes, however, most of the emerging markets have more fixed exchange rate regimes. This difference may partially explain the reasons why there exists negative relationship between exchange rate volatility and net debt issuance in emerging markets. Our results suggest that there is weak evidence in support of the existence of the exchange rate risks associated with NFC international debt issuance in developed economies. However, it does not imply that there is no financial risk related to this behavior. Our results indicate that there may exist a transmission channel through international debt issuance to spread the domestic stock market risks to other countries.

In Table 4, we clearly see that the coefficient on stock market growth rate is positive and significant across all six model specifications. It is slightly surprising to find out that NFCs' overseas debt issuance is strongly correlated with the domestic stock market growth rates. Normally corporate governance textbooks suggest that NFCs should focus on their main business operation and should not speculate in financial market. Interestingly, using emerging market data, we see the coefficients of stock growth on net debt issuance are insignificant across different models. Even though the coefficient is still positive and marginally significant at 10 percent level when we include the whole set of year dummies, the magnitudes of these coefficients are much smaller compared to developed economies.

Generally speaking, developed countries on average have less international capital control compared with emerging markets. Many emerging markets have laws to prevent the international capital to speculate in the domestic asset market. If NFCs want to speculate in the stock market to make momentum profit, it is not optimal to speculate in the countries that create barriers to conduct carry trade. From micro perspective, NFCs firms headquartered in the developed countries may have much larger market capitalization and have diversified business operations, compared to the counterpart who issues the debt overseas in emerging markets. For example, many NFCs in devel-

oped countries issue credit cards to help their costumers to finance their purchases. Therefore, if these companies are classified as NFCs, they may not behave as the textbooks suggest. Generally speaking, we suspect that in the developed countries, many NFCs may have also provided financial services related to their products, which makes them different from the NFCs in traditional concepts. They may borrow cheap from overseas and invest in the stock market like other domestic financial institutions.

To see the potential stock market risks associated with NFC overseas borrowing, we use Hansen (1999) approach to check if there is threshold effect of stock growth rate on international debt issuance, i.e., whether the impact of stock market growth differs when it is below or above a certain threshold. The results suggest, there may exist threshold effect of this impact around 7-percent quarterly stock growth rate.

Inspired by this result, we run original panel regressions with observations below and above this threshold separately. From Table 6, we can see when the stock market growth rate is less than 7 percent, the coefficient of stock market growth on net debt issuance is much larger than when the stock market growth rate is more than 7 percent. The finance literature suggests that the expected average annual stock return is around 10 percent. If 7 percent is the threshold point, it implies that if the annual stock growth rate is above 30%, the larger the stock market growth rate, the lower the net international debt issuance.

This is a surprising result as it suggests that NFCs have less incentive to borrow money from overseas when the stock market is booming, given other things equal. The structural break may be largely due to the supply side effect, i.e., international investors may be cautious about the financial risks in the local economy. If the stock market bubble bursts, then it is possible that both NFCs and domestic financial institutions may not have enough foreign liquidity to pay for the international debts. However, the harder NFCs borrow money to roll over the existing debts, the higher the possibility to default on these debts. The debt rationing behavior, though it is rational at individual level for international investors, can lead to the undesired outcomes collectively in the global financial market.

4.2 NFC's incentives to issue debt overseas

There are some potential incentives for NFCs to issue debts overseas. It is common for exporters to issue debts in foreign currencies to offset the exchange rate risks in the incoming foreign currency income. The interest rate gap can also be a potential motivation for NFCs to borrow overseas to lower the cost of capital. In this study, we test these hypotheses by adding current account balance and interest gap as regressors in the panel regressions. Other than these international factors, we also include domestic GDP growth rate to control for the fundamentals of domestic economies.

As suggested by many researchers in the international economics literature, changes in current account may have a positive and significant impact on net debt issuance. This behavior is explained by the hedging of the foreign currency revenue of the exporting firms. Our results support this hypothesis. This result is robust across the all model specifications using the full sample as well as the observations from the developing countries. However, this coefficient is negative and insignificant in case of developed economies. Why do NFCs headquartered in the developed countries not hedge against their foreign currency incomes? One possible explanation is NFCs headquartered in the developed countries may have more foreign direct investment compared to the emerging market. If these NFCs have factories overseas, they need to pay the local employees in local currencies. The FDI from developed countries to emerging market may explain part of the reasons why there exist lower degree of currency mismatch in the NFCs headquartered in the developed countries. In contrast, NFCs headquartered in the emerging market may only have foreign sales but not foreign operation costs, therefore these exporters need to hedge the exchange rate risks through issuing debt securities overseas.

In our results, NFCs do not seem to issue debt abroad to finance domestic investment projects. The coefficients of GDP growth rate on net issuance are insignificant among all the regressions. It is not surprising to find this result because, controlling for interest rate gap, if NFCs need to finance an investment project, it is more convenient for them to raise money domestically to avoid the extra cost from currency conversion and other related costs in overseas issuance.

The interest rate gap between domestic and international debt markets plays an insignificant role in explaining the overseas debt issuance in our results. The interest rate, the price of using capital, is an essential factor to direct capital flow. The interest rate is jointly determined by the supply and demand in the debt market. The insignificant coefficient in this regression is probably due the opposite effects from supply and demand sides. For NFCs, it may be attractive to borrow from overseas at a much lower interest cost; however, for international investors, the large interest gap may imply that there are potentially other financial risks to explain why the domestic interest rates are much higher than the international interest rates. As international investors, they face information asymmetry in the financial market compared to the local financial institutions, so they may have concern about the adverse selection in the international debt market. The risks associated with purchasing international debt securities may dominate the interest rate gap in the global financial market.

5 Conclusions

This paper studies the behavior of the non-financial firms (NFC) overseas debt issuance, using a newly developed dataset on international debt securities by the Bank of International Settlement. Our study used the panel dataset of 32 countries for the sample period 1993-2015. The results suggest that, the depreciation of domestic currency against US dollar is negatively correlated with the international debt issuance. Domestic stock market growth rate is positively correlated with international debt issuance behavior in developed countries. In developing countries, exchange rate volatility and changes in current account play a significant role in international debt issuance. These results seem to be consistent with carry trade hypothesis and suggest the potential for exchange rate risk in debt rollover.

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Table 1: Data Sources

Variable	Data Source
Net Debt Issuance	BIS
Exchange Rate	FRED, OANDA
Stock Market Index	FRED, WSJ
Real GDP	FRED, IFS
Federal Fund Rate	FRED
Discount Rate	FRED and Central Bank websites
Current Account Balance	IFS

Table 2: Summary Statistics

Variable	Mean	Std. Dev.	Min.	Max.	N
Net debt issuance(billions of US dollar)	0.358	2.077	-22.935	23.02	2720
Exchange rate volatility	2.82	35.909	0	1151.724	2738
Exchange rate	356.624	1599.789	0.014	10000	2738
Stock growth rate(%)	0.034	0.152	-0.712	2.918	2683
GDP growth rate(%)	0.009	0.018	-0.121	0.423	2637
Interest gap(%)	5.512	11.856	-6.02	194.2	2641
Current account difference(billions of US dollar)	16449.863	2054232.385	-35358512	42763492	2571

Table 3: The Determinants of International Debt Issuance

	(1)	(2)	(3)	(4)	(5)	(6)
	debt_dif	debt_dif	debt_dif	debt_dif	debt_dif	debt_dif
Exchange rate volatility	-0.857*** (-7.87)	-0.966*** (-8.71)	-1.016*** (-6.95)	-1.015*** (-6.47)	-1.013*** (-6.39)	-0.871*** (-3.11)
Exchange rate	-0.0224 (-0.83)	-0.136** (-2.50)	-0.145** (-2.44)	-0.160** (-2.72)	-0.164** (-2.74)	-0.363* (-1.77)
Stock growth rate		736.3** (2.28)	758.4** (2.35)	825.5** (2.41)	853.7** (2.40)	658.2** (2.68)
GDP growth rate			-802.0 (-0.58)	-1372.5 (-0.97)	-1524.0 (-1.06)	130.5 (0.07)
Interest gap				-8.804 (-0.95)	-8.758 (-0.93)	12.55 (0.90)
Current account difference					9.15e-09*** (49.38)	9.80e-09*** (5.75)
Year-fixed effect	No	No	No	No	No	Yes
Country-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Constant	376.6*** (37.51)	404.9*** (21.81)	420.1*** (16.03)	485.7*** (11.04)	502.4*** (11.20)	-540.9 (-1.08)
<i>N</i>	2655	2562	2511	2393	2258	2258

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: The Determinants of International Debt Issuance (Developed Countries)

	(1)	(2)	(3)	(4)	(5)	(6)
	debt_dif	debt_dif	debt_dif	debt_dif	debt_dif	debt_dif
Exchange rate volatility	-66.49 (-1.09)	43.84 (0.35)	37.61 (0.29)	-82.43 (-0.50)	-81.42 (-0.47)	-26.38 (-0.19)
Exchange rate	-15.59 (-1.02)	-15.47 (-1.02)	-17.84 (-1.12)	-14.59 (-1.56)	-14.34 (-1.60)	-19.50*** (-4.50)
Stock growth rate		2504.6* (1.85)	2693.6* (2.05)	3277.9** (2.67)	3486.4** (2.67)	2645.7** (2.52)
GDP growth rate			-5969.0 (-1.21)	-4124.8 (-0.87)	-3961.0 (-0.82)	652.7 (0.15)
Interest gap				158.2 (1.10)	167.3 (1.10)	97.36 (0.99)
Currenct account difference					-0.0302 (-1.09)	-0.0170 (-0.89)
Year-fixed effect	No	No	No	No	No	Yes
Country-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Constant	867.4** (2.80)	791.5** (2.34)	881.7** (2.46)	703.2*** (4.63)	709.4*** (4.79)	-589.0 (-0.86)
<i>N</i>	1012	992	978	976	926	926

t statistics in parentheses* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: The Determinants of International Debt Issuance (Developing Countries)

	(1)	(2)	(3)	(4)	(5)	(6)
	debt_dif	debt_dif	debt_dif	debt_dif	debt_dif	debt_dif
Exchange rate volatility	-0.841*** (-8.64)	-0.934*** (-8.78)	-0.918*** (-7.43)	-0.915*** (-7.10)	-0.911*** (-7.01)	-0.695*** (-6.15)
Exchange rate	-0.0212 (-0.79)	-0.0824** (-2.43)	-0.0813** (-2.25)	-0.103** (-2.66)	-0.105** (-2.70)	-0.0727** (-2.74)
Stock growth rate		200.7 (1.31)	189.3 (1.20)	246.0 (1.66)	261.2 (1.72)	319.6* (2.06)
GDP growth rate			349.8 (0.34)	-430.7 (-0.56)	-541.4 (-0.68)	-149.0 (-0.11)
Interest gap				-13.84* (-1.90)	-13.80* (-1.87)	-4.444 (-0.60)
Current account difference					8.97e-09*** (67.73)	1.04e-08*** (8.54)
Year-fixed effect	No	No	No	No	No	Yes
Country-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Constant	269.2*** (17.16)	304.2*** (16.89)	300.0*** (11.86)	437.7*** (7.89)	445.2*** (7.96)	390.9 (1.37)
<i>N</i>	1643	1570	1533	1417	1332	1332

t statistics in parentheses* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: The Threshold Effect of Stock Growth on International Debt Issuance

	(1)	(2)
	debt_dif	debt_dif
Exchange rate volatility	-18.48 (-1.40)	-0.814*** (-3.08)
Exchange rate	-0.450 (-0.36)	-0.375** (-2.27)
GDP growth rate	-2490.1 (-0.72)	737.8 (0.34)
Interest gap	20.18 (1.50)	12.38 (0.79)
Current account difference	4.93e-08*** (7.75)	-1.59e-08*** (-9.05)
Stock growth rate	724.0* (1.72)	1724.0* (1.93)
Year-fixed effect	Yes	Yes
Country-fixed effect	Yes	Yes
Constant	722.3 (0.72)	-1575.2 (-1.68)
<i>N</i>	692	1566

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$